

# GTIN Scanner

A handheld scanning app that replaced a legacy Excel macro for supply-chain receiving — built for the pace and unreliability of a real warehouse floor, not a desk.

**Prepared for** Directors & stakeholders      **Status** In production

**Data source** Microsoft Fabric Lakehouse (live since Aug 7, 2026)      **Updated** Aug 31, 2026

## 01 · OVERVIEW

### What it replaces, and why it exists

Pickers used to scan barcodes into an Excel macro to check items against contract lines. It worked, but it lived on one machine, had no audit trail, and gave supervisors no visibility into what was happening on the floor in real time.

The GTIN Scanner is a web app pickers open on a handheld device. They scan a badge to start a session, then scan items — the app looks each one up instantly against the current supply contract and flags anything on hold. Supervisors watch every active session from a separate monitor board and can export or close any of them.

**184,411**

contract lines indexed for  
instant lookup

**O(1)**

lookup time per scan —  
dictionary, not a search

**3 days**

session data retention window  
(export before then)

## Two screens, one shared record

Every session and scan writes to the same Postgres database, so the handheld and the monitor board are always looking at the same truth — there's no sync delay and nothing lives only in one browser tab.

### PICKER

#### Handheld

Scans a badge and a location to start a session, then scans items in a loop. Only the device that started a session can end it normally.

### SUPERVISOR

#### Monitor board

A live list of today's sessions — who, where, how many scans, last activity — with drill-in, per-session export, and a Force End for a device that dropped off.

### MANAGER

#### Admin page

Email-allowlisted refresh control for pulling the latest contract data on demand, plus the same Force End control.

# Local first, internet second

This is the core reliability decision in the app: the contract catalog is loaded into memory once, so **the large majority of scans never touch the network at all**. The internet is only involved for the rare item that isn't on the current contract.

|                                                                                  |                                                                                                       |                                                                                             |                                                                                       |
|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <div>1 · SCAN</div> <div>Barcode read, GS1 wrapper parsed, GTIN extracted.</div> | <div>2 · LOCAL LOOKUP</div> <div>Checked against the in-memory contract index. No network call.</div> | <div>3A · HIT</div> <div>Item, on-hold status, and unit-of-measure shown immediately.</div> | <div>3B · MISS</div> <div>Falls back to the FDA's public device-identifier API.</div> |
|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|

If that fallback call also fails — a timeout, a bad connection — the app records it as a clean "Not Found" instead of erroring out, and logs *why* it missed (wrong check digit, not on the current contract, inactive line, etc.) so the miss is explainable later, not just a blank.

| EXAMPLE SCAN   | ITEM                  | RESULT                         |
|----------------|-----------------------|--------------------------------|
| 00841098765432 | Surgical Mesh 15cm    | Found — contract line          |
| 00123456789012 | Nitrile Gloves, Large | Found — <b>on hold</b> warning |
| 00000000000000 | Not on contract       | FDA API fallback               |

## Built for the floor, not the demo

Every feature below exists because of something that broke, stalled, or annoyed someone during real use — not a hypothetical.

### Hands-free auto-submit

SPEED

#### PROBLEM

The warehouse's actual scanners (Zebra handhelds) deliver a whole barcode in one instant burst, not as typed keystrokes. The first version of auto-submit was tuned for a keystroke-by-keystroke scanner and simply never fired on the real hardware — pickers had to tap the screen after every item.

#### SOLUTION

Two independent triggers now cover both delivery styles: scan-speed cadence for guns that type character-by-character, and a check-digit validity test as a backstop for guns that paste the whole code at once. Manual typing still works — Enter or the Look Up button always did.

### Audible confirmation

SPEED

#### PROBLEM

Pickers are watching the item in their hands, not the screen. A quiet or ambiguous tone was easy to miss over warehouse noise, or read as a glitch rather than a confirmation.

#### SOLUTION

A distinct two-note chime on every successful scan, with separate alarm tones for on-hold and not-found results — loud and clear enough to register without looking up, and unmistakably different from an error.

## Session resume after a dropped connection

RELIABILITY

### PROBLEM

Warehouse wifi is inconsistent, and mobile browsers routinely kill or reload a backgrounded tab. Losing the tab used to mean losing the session and starting over.

### SOLUTION

Every session and scan is written straight to the database, not just held in the browser tab. The session ID lives in the page URL, so a refresh, a killed tab, or a full browser restart reopens the same session with its full scan history intact — nothing is lost, and no scan has to be redone.

## Graceful network fallback

RELIABILITY

### PROBLEM

The fallback lookup depends on an external, public FDA database over the internet — a connection that isn't always available or fast on a warehouse floor.

### SOLUTION

A network failure there never crashes the scan — it resolves to a clean "Not Found" with the reason logged, so the picker keeps moving and the miss is still traceable afterward. And because most scans hit the local index first, this fallback is rarely exercised at all.

## Fast cold start

SPEED

### PROBLEM

Loading the full 184k-row contract index used to happen before the app showed anything, adding a multi-second blank wait every time the app woke up.

### SOLUTION

The index now loads lazily, in the background, after the picker's start screen is already on screen — visible progress instead of a blank wait, with no change to how fast lookups run once scanning starts.

## On-hold and duplicate-scan awareness

DATA QUALITY

### PROBLEM

A found item isn't automatically a safe item, and rescanning the same barcode twice used to either get silently dropped or inflate the count as if it were a new item.

### SOLUTION

Items flagged on hold in the contract data show a distinct warning instead of a plain match. A repeat scan of something already in this session is recognized instantly (no re-lookup) and its count increments, so exports reflect exactly what was physically scanned.

## Explainable misses

DATA QUALITY

### PROBLEM

A blank "Not Found" gives a supervisor nothing to act on — is the item genuinely new, mislabeled, or just off-contract?

### SOLUTION

Every miss is diagnosed and tagged with a specific reason (bad check digit, not on the current contract, inactive line, etc.), stored with the scan and surfaced in both the app and the Excel export.

## 05 · DATA & RETENTION

# What's stored, and for how long

Sessions and scans are kept for **3 days**, then purged automatically by a daily scheduled job — this runs independently of the app, so it fires even if nobody has the app open. Export any session you need to keep before then; there's no longer-term archive in this version.

Each session can be exported to a single Excel workbook from the monitor board — one file per session, named with the location, picker, and date so a batch of exports stays sortable.

## Who can do what

Anyone can pick up a device and start a scanning session — that's the front-line workflow and it's intentionally open. Two actions are restricted to a named allowlist of managers and supervisors: refreshing the contract data on demand, and force-ending a session that a device dropped without properly closing.

**Note on the access model:** admin access today is a typed-email allowlist check, not a verified login — closer to a sign-in sheet than a badge reader. Every attempt, granted or denied, is logged, so misuse is visible after the fact even though it isn't blocked up front. This is a known, deliberate trade-off for now; see Open Items.

## What's next

- **Verified admin login.** Replace the typed-email allowlist with real sign-in through the company's existing Microsoft identity — same provider already used for the contract-data connection, so no new system to stand up.
- **Deployment sign-in for the live data connection.** The last piece of the Fabric lakehouse cutover is wiring up unattended, headless authentication for the hosted app, rather than the interactive sign-in used in development.
- **Web app team hand-off.** Looping in the broader web app development team as the next step for longer-term ownership.